

## Introduction

Provided with a dataset on hospital specimen journeys, our key problem is visualizing this complex and large dataset as an average time series that allows our clients and stakeholders (hospital staff, lab operations and quality assurance staff) to view the time taken between stages such as from ordering a test to collecting it, from collecting the test to receiving it, and from receiving the test to verification. In addition, our goal is to support A/B comparisons among timelines, allowing staff to view differences and characterize test taking workflows in side by side comparisons. In order to quantify risk and where and in what conditions bottlenecks could occur, we also calculated cancellation rates among different A/B cohorts.

From this problem, we formulated our dashboard as a solution to these requirements for the client. Our dashboard features an A/B comparison feature, allowing users to compare timelines and average cancellation times. The default is set towards comparing weekend vs weekday timelines and cancellation rates, but supports comparisons between event street, department, and test code, that are easily configurable through a sidebar. To better support staff in narrowing down specific tests, we added filtering features based on test codes, event street, and performing departments, as well as risk likelihood for cancellations and likelihood of long test stages.

This problem and its approach matters because without visualizing bottlenecks and cancellation risks, as well as the tradeoff between these event risks and the speed of test completion become opaque, making it less likely for hospital staff to be able to identify areas to improve the test collection workflow. We hope that our dashboard will allow hospital staff to easily identify these bottlenecks, assess tradeoffs in current workflows between speed and cancellation risk/longer specific stages in the pipeline, and make actions according to the data.

## Exploratory Data Analysis

Our first step in approaching the problem was to understand the data, the individual rows, and the fields. Looking through the data, we noted that each row was an event, not a complete order from start to finish. We noted that **test\_performing\_dept** and **event\_street** are important fields to include as metadata for eventual filtering or A/B analysis. We note the unique identifiers that are available that we can use to track events and order stages, such as **accession\_id**, **barcode** and **tube\_id**, and **specimen\_id**. We also take note of **event\_source** and decide to focus on **event\_source=order**, as the client request and problem is primarily focused on a timeline of order completions.

With an understanding of the metadata, we noted fields specific to what would be essential in visualizing our time series data. Specifically, we found **test\_ordered\_dt**, **test\_collected\_dt**, **test\_receipt\_dt**,

**test\_min\_resulted\_dt/test\_max\_resulted\_dt**, and **test\_min\_verified\_dt/test\_max\_verified\_dt**. These fields would be crucial as they gave us the time a test was ordered, collected, received, and verified, allowing us to calculate average times between each of these stages to visualize as an average time series.

With all of these fields, we finally wrote our data processing script. Our core logic is to create a global unique identifier with **accession\_id + test\_code**, and calculate the time it takes between each stage for this ordered test. Other data such as day of the week and whether the test was cancelled was added as a field on our processed data.

With our initial data processed but without any dashboard visualization yet, we calculated average and median hours for each ordered test stage, as well as p90 hours:

```
cohort,stage,available_records,median_hours,mean_hours,p90_hours
Weekday,order_to_collect,152932,0.167,0.616,1.45
Weekday,collect_to_receipt,186464,0.433,3.359,6.867
Weekday,receipt_to_verified,179146,0.583,7.662,5.983
Weekday,order_to_verified,180758,1.75,9.353,9.517
Weekend,order_to_collect,29077,0.383,0.875,1.867
Weekend,collect_to_receipt,37923,0.267,1.858,3.833
Weekend,receipt_to_verified,36264,0.483,2.497,1.417
Weekend,order_to_verified,36700,1.1,4.303,6.317

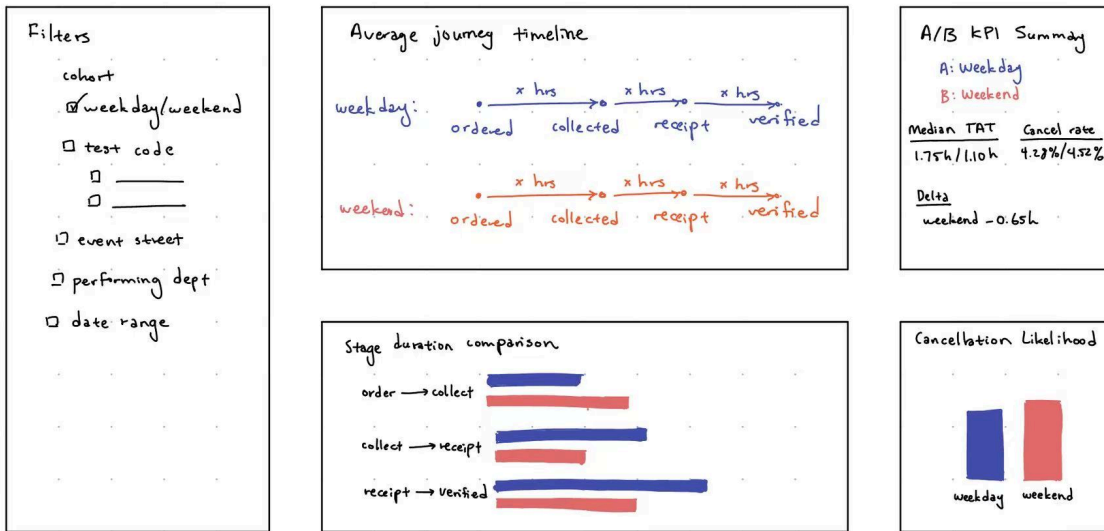
cohort,ordered_tests,completed_tests,cancelled_tests,cancellation_rate
Weekday,191460,180776,8195,0.0428,0.167,0.433,0.583,1.75
Weekend,38785,36708,1754,0.0452,0.383,0.267,0.483,1.1
```

(The image cuts off, but all the rows are: cohort, ordered\_tests, completed\_tests, cancelled\_tests, cancellation\_rate, median\_hours\_order\_to\_collect, median\_hours\_collect\_to\_receipt, median\_hours\_receipt\_to\_verified, median\_hours\_order\_to\_verified)

These initial results were more to aide our exploration of initial low-fidelity sketches of our dashboard, which we made to get a better idea of how we wanted to layout our dashboard and use graphs and encodings that best supported our desired tasks: to compare across A/B cohorts, identify bottlenecks, and find areas of friction/longer times between stages or higher cancellation risk.

From that, we drew these prototypes to consider different ways of encoding our information to best fit our tasks:

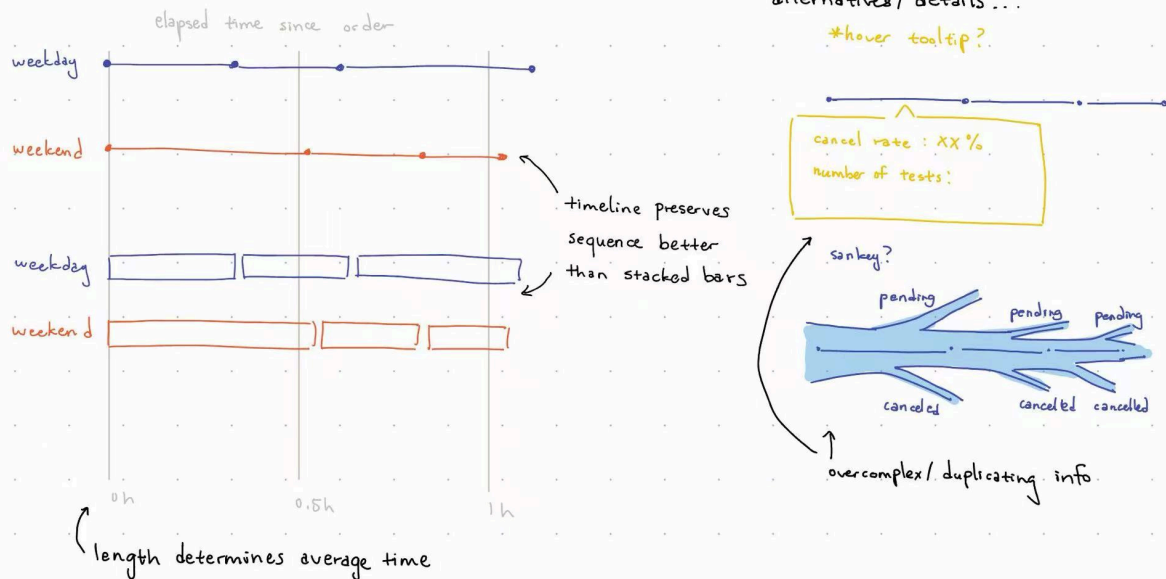
## Sketch 1: Dashboard Overview



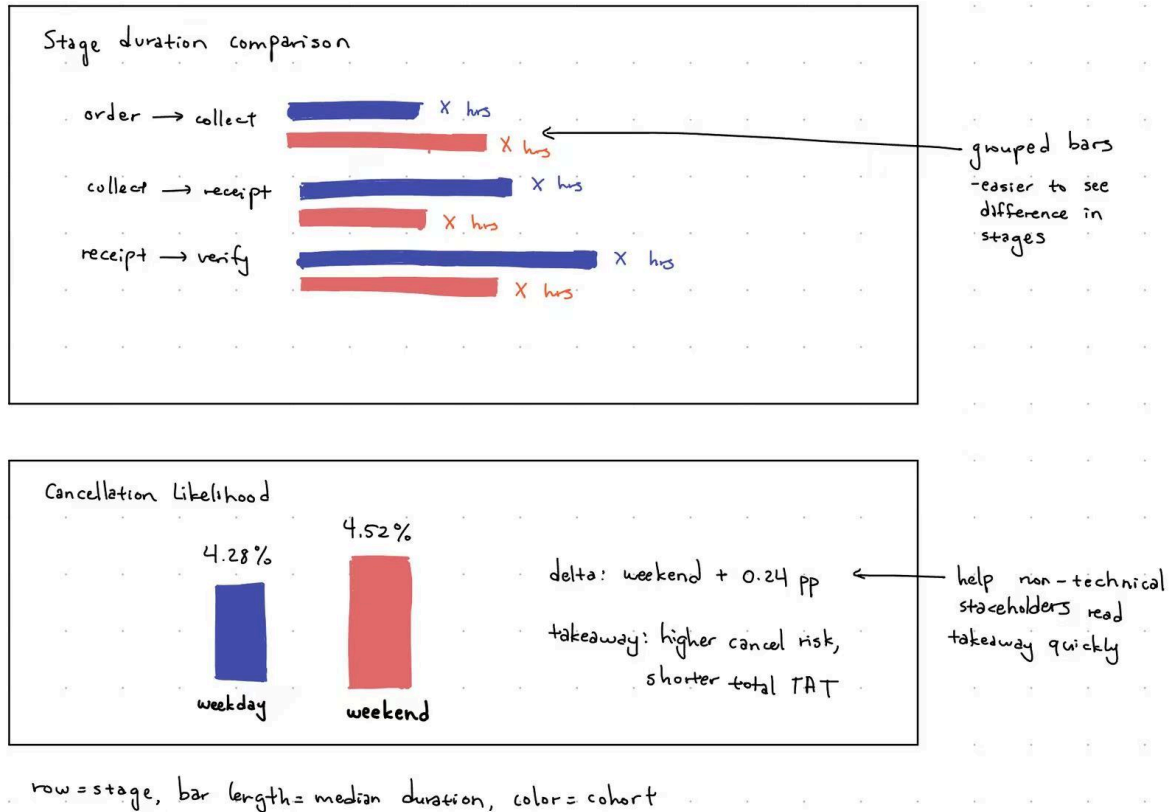
### Notes:

- use color for cohort
- use x-position and bar length for time
- timeline should be centerpiece
- cancellation separate from timing so risk not confused w/ duration

## Sketch 2: Average Journey Timeline options



### Sketch 3: A/B Comparison and Event Likelihood



Most notably, we settled on a side by side timeline as the main centerpiece of the dashboard. Then, we decided to include group bars to compare stage durations and cancellation likelihoods for easy A/B comparison and analysis. From this we were able to settle on what charts we would use, as well as color encoding and layout.

### Analysis and Visualization Design

We were able to base our final dashboard design on our initial sketches, refining our initial design based on feedback, but retaining things we liked, such as a clear A/B color encoding and the basic average timeline comparison structure.

One of the things we iterated on based on feedback was an overall view at the A/B comparison at the top of the dashboard. In order to make insights explicit so stakeholders can easily compare cancellation risks and median ordered test completion times between the two cohorts, we added a summary to the top of the dashboard and notes for non technical users to understand key data insights.

## Specimen Journey Comparison Dashboard

Compare average specimen timelines, bottleneck stages, and event likelihood across mutually exclusive cohorts. The current lens is **Weekpart**, with A = **Weekday** and B = **Weekend**. Current event definition: **Cancellation** (``cancellation_dt``).

Cohort A sample

**191,440**

ordered tests after filters

Cohort B sample

**38,785**

ordered tests after filters

Cancellation delta

**+0.24 pp**

B minus A cancellation rate

Median TAT delta

**-0.65 h**

B minus A order-to-verified median

### CURRENT READ

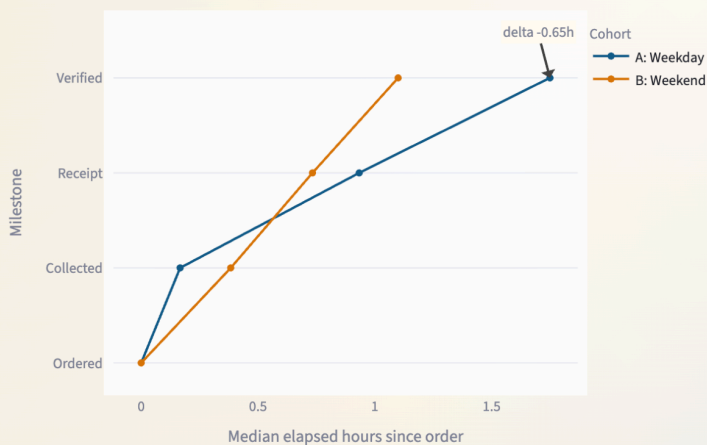
- Cohort B is 0.65 hours faster than Cohort A on median order-to-verified turnaround time.
- Cohort B changes cancellation likelihood by +0.24 percentage points relative to Cohort A.
- The largest stage gap appears in Order -> Collect, where Cohort B is 0.22 hours slower.

Below our summarization, we have our average journey timeline, a line graph where the x-axis represents the median hours elapsed since each order stage, with each stage as expressed in order on the y-axis as ordinal attributes. We encoded each cohort with a different color to clearly express the differences in each timeline. We felt that a timeline/linechart was one of the clearest ways to express and compare time elapsed, as it allowed for easy and intuitive comparison of lengths as well as a clear view of the average order timeline as it passed through collection, receipt, and verification.

In addition, we added interpretation and A/B snapshots to give supplemental information on the amount of tests. The interpretation hints help clearly delineate what is the largest stage gap, allowing users to identify bottlenecks in the process as orders move toward completion.

**Overview** Hotspots Details & Method

### Average Journey Timeline



Each point shows the median arrival time for a milestone, relative to the ordered timestamp.

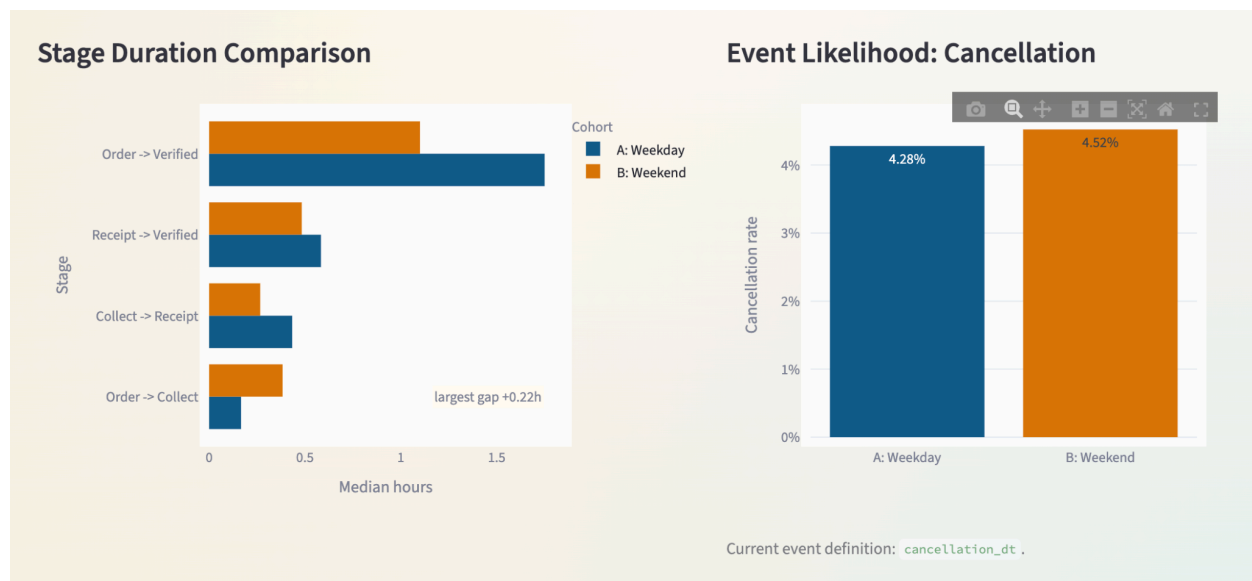
### A/B Snapshot

| cohort     | ordered_tests | completed_tests | Cancellatio |
|------------|---------------|-----------------|-------------|
| A: Weekday | 191,440       | 180,760         |             |
| B: Weekend | 38,785        | 36,708          |             |

### Interpretation Hints

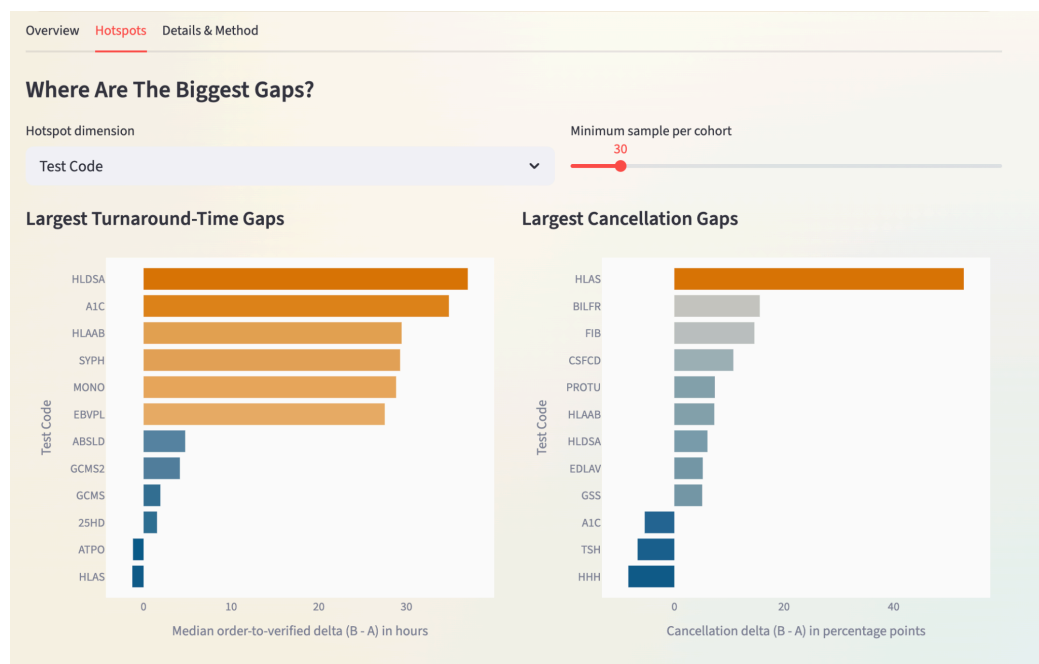
- Largest current stage gap: **Order -> Collect**
- Cohort A median TAT: **1.75 h**
- Cohort B median TAT: **1.10 h**
- Cancellation delta: **+0.24 pp**

In order to aid in understanding event likelihoods such as cancellation risk, below the main timeline we added side by side event cancellation likelihoods for A/B cohorts. To understand which stages take the longest as well, we added grouped bar charts for stage duration, also comparing them via A/B cohorts. We felt that simple bars here were the best encoding for easily understanding and comparing time or comparing risk likelihood.



Lastly, in order to support comparison between cohorts and identify the largest gaps between the cohorts, we supported a hotspot view that directly showed the magnitude of difference in order completion time. This helped identify the largest gaps across another variable/dimension, such as test code, event street, performing department, or day of the week.

By encoding this in a multidirectional bar chart, it made it clear and easy to see which direction the difference would be - whether it took more time for cohort A or cohort B in this particular dimension, with length being an easy way to identify the magnitude of difference.



To supplement our main dashboard to provide a range of A/B comparisons and filtering options to narrow down the scope of these A/B comparisons, we added a sidebar that allows the user to filter by test code, event street, performing department, and choose the type of A/B comparison to show within the main dashboard.

This supports our goals of making the dashboard a way to compare across a range of A/B cohorts and identify bottlenecks across a range of variables and dimensions. We hope that by providing this it allows exploration of the data to be more extensible and insightful.

## Insights and Discussion

From our own exploration of the data and our dashboard tool, we were able to find some interesting insights. We focused our main dashboard and timeline on the comparison of weekdays vs weekends. From this, we found that weekends had an overall faster turnaround time for orders, with an average time of 1.10 hours vs the weekday's 1.75 hours.

However, what was interesting to note is that this did not necessarily indicate that the timeline was smoother across all orders on weekends. In particular, we found that weekends had a higher risk of long order-to-collect delays, with a 9.6% chance of a longer time

before collection for weekends versus a 7.6% chance on weekdays. Cancellations were also slightly higher on weekends, with a 4.52% cancellation risk versus weekday having a 4.28% cancellation risk.

While we can't conclude that these differences are causal as our dashboard purely reports on data and can't prove causation for these correlations, we can still consider whether these higher percent risks are a tradeoff for overall lower order completion times. Hospital staff aided with this data and an understanding of their workflow can consider why these tradeoffs could be occurring and whether they think they are related.

One last observation we made through exploring our dashboard was an interesting anomaly - we considered event street as a variable to make A/B comparisons and found that Taubman had high risks and common delays in the receipt to verification step of the order pipeline. In particular, Taubman had a 40.67% risk of long receipt to verification step compared to other streets, such as Medical, with 8.53% and Hospital, with 7.19%. From this anomaly we found that our dashboard could also be useful for identifying outliers and particular types of orders that could take especially long in certain stages of order completion.

## **Conclusion**

Overall, from analyzing the dataset and building the dashboard we found that time-series data is complex and can be affected by a range of variables, especially with this dataset, where things like event street, performing department, and day of the week could change our average timeline. We found that our dashboard could be useful at identifying tradeoffs between risk and speed, as we were able to visualize the overall order completion speed between two cohorts as well as the event likelihood risks of cancellation or delays in order stages, all in the same page.

However, there are still limitations to our current dashboard. In terms of data exploration, cancellation risk is mostly viewed as an aggregate, which means whether cancellations happen more frequently across one stage or another is insight that can't be acquired through this dashboard. Another of our gaps in analyzing the data was tube tracker movement, as we focused on order level data for our dashboard. Overall, these are the gaps that we found due to focusing on an overview level of larger trends in cancellation/ long order risk, and visualizing average timelines over specific events. However, we think that specific data such as tube tracker events would be useful and could be integrated into the dashboard as a next step.

Overall, we were still able to draw some interesting insights from exploring our dashboard despite these limitations. Aggregating our time series data using the dashboard allowed us to explore these trends across multiple dimensions and find potential tradeoffs between order completion speed and cancellation/early delay risk.